

An object is a code that contains data or variables with the relevant methods or functions that perform actions on that data giving a particular output. A class, is the layout or definition of that object. Think of the final headline problem that needs to be solved. This is the final object that is to be created, say a car. To construct this car, you need several small pieces or parts. Every part is also an object. The basic plans drawn up outlining these parts is the class. Once all the smaller parts or objects are created, they can be put together to make the final object, the car. This, in a nutshell, is object oriented programming.

Many objects can be created from a class. Instance refers to an individual object of a certain class. E.g. object *obj* that belongs to a class *Circle*, for example, is an instance of the class *Circle*. The process of creation of an instance of a class is called instantiation. Another similar feature is a class variable. They are defined within a class and are shared by all instances of a class. Both, instance variable and class variable are part of data members which holds all the data associated with the class. Method is a special type of function defined within the class.

Syntax

A class is defined using the keyword *class* followed by the docstring or the class description given in triple inverted quotes. The docstring is optional but recommended for easier understanding of the larger code at a later stage. Finally, you end it by the class suite which can be lines of code in itself. Let's construct a repository of employee names, id's and locations.

First we define and create the class as shown here: <https://digit.in/ClassPy>. Breaking it down, the first variable created is the `empCount`. This is a variable created for the class `Employee` and can be accessed from inside or outside the class. Being a class variable means that its value is shared amongst all instances in this class. Once the variable is defined, the various functions are given which tells Python what data to pull up when called. In a class function or method, the first argument is always self. The first function is `__init__()` which is a special method. It is also called as class constructor or initialization method that Python calls when you create a new instance. There are 3 variables that are to be displayed- `name`, `employee id` and `location`. So in parentheses, after the mandatory self argument, the other arguments or input parameters are given as name, employee id and location. Next, the arguments have to be defined in terms of self using the dot notation so every argument will be defined as `self.arg = arg`. The next